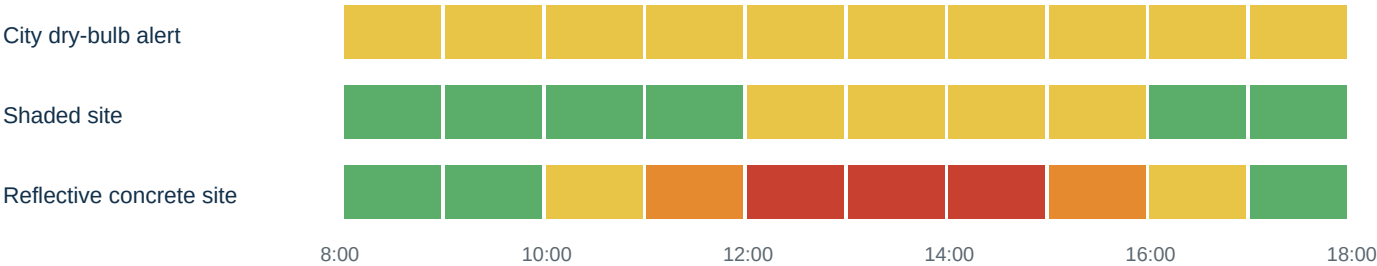


ShiftShield

Final plan: the heat alert that proves the rest happened

A shift-level heat safety agent for a single worksite. Site-specific WBGT, a work-rest schedule and an alert an hour early, and then the part no existing tool has: a record of the break confirmed by the supervisor and, anonymously, by the workers themselves.



Illustration, not data: the same city and the same hours. A city dry-bulb alert gives one answer for everyone; a site-level estimate separates a shaded site from a reflective concrete one. Green is normal work, then caution, high and very high.

Item	Detail
Hackathon	Environmental Hacks, WeMakeDevs x AWS. Online Oct 8 to 11, 2026. Optional build day at DTU, Delhi, Oct 10.
Track	Track 02, Heat and Water. The track lists heatwaves explicitly.
Problem source	Problem statement supplied by the team's Head of Department (full text in Section 3).
Team	EcoNexus, team code AVYAD9. Leader: Sameer Ahmed. 2 of 4 seats filled when version 1 was written.
This version	Final plan (version 7), prepared October 5, 2026. Adds the heat action plan rulebook agent and the single-product framing. Replaces versions 1 to 6. Planned for two people, with notes for four.
Prizes	Track winner: Rs 2,00,000 cash plus \$2,000 AWS credits. Four runner-up teams: \$1,000 credits. Top 5 blogs: AirPods. Up to 10 students across tracks may be put forward for fast-track Amazon interviews.

Read first

This is a planning document. The rules say old projects do not count, and a repository whose history does not match the event dates disqualifies the whole team. This document contains design, formulas and short pseudocode only. The repository is created on Thursday, Oct 8.

SECTION 1 | NEW IN V3

The gap: what exists, and what we build that does not

A market check done on October 3, 2026. It is a search, not a patent review, so the claim is 'we found no tool that does this'.

Existing tool	What it already does	What it does not do
Perry Weather, Kepler 51 (US, commercial)	On-site sensors, WBGT forecast, automatic alerts with work-rest instructions, compliance logs.	Needs hardware and an enterprise budget. The log records that an alert was sent, not that workers rested.
AIHA Heat Stress app	Free WBGT estimate from location, work intensity, clothing and sun exposure; rest and water advice.	No Indian languages, no site record, nothing from the workers' side.
OSHA-NIOSH Heat Safety Tool	Hourly risk level and precautions for a location.	Uses heat index, so it ignores sun and radiant heat.
SHRAM dashboard (UC Berkeley, India)	Heat stress forecasts and alert sign-ups across India.	District level. No worksite, no schedule, no record.
Delivery platform apps (India)	Weather alerts and SOS support inside rider apps.	Advice only. Nobody checks whether rest points are usable.
SEWA parametric heat insurance	Automatic payouts to about 50,000 women workers.	Triggered by city air temperature, not site conditions or hours actually lost.

Three gaps we found

- **Nobody records whether the rest happened.** Delhi's Heat Wave Action Plan 2026 is reported to mandate rest from 1 PM to 4 PM for all outdoor workers, and the coverage names no way to record compliance. Researchers quoted on the NDMA advisory for gig workers say there is no feedback loop from workers.
- **A fixed window is a blunt rule.** A blanket 1 to 4 PM break can miss a dangerous humid morning. No tool shows, for one site, which dangerous hours fall outside the mandated window.
- **Nobody counts the hours lost to heat at a site.** Daily-wage workers lose pay when work stops, and the one insurance scheme that pays for heat uses city air temperature.
- **Nobody tells a supervisor which obligations apply to their site today.** Heat action plans are long documents that differ by city and state. A Centre for Policy Research review of 37 plans found no national repository and only a few plans easy to find online.

What ShiftShield claims

ShiftShield is a hardware-free heat safety agent for one Indian worksite that closes the loop: it tells the supervisor what to do an hour ahead, then keeps a rest record confirmed from both sides, checks the site against the fixed-window rule, and totals the work hours lost to heat. Say it this way to judges: "Others send the warning. We record whether the rest was given, and the workers are the ones who confirm it."

One product in place of several tools

What a supervisor needs	What it takes today	In ShiftShield
A heat warning	City SMS or a district dashboard	A site alert, an hour early, by text and voice
A heat estimate and rest advice	A separate app	Site WBGT and a work-rest schedule, built in
A compliance log	Paid products with on-site sensors	A rest record with no hardware
Which rules apply today	Reading the heat action plan yourself	A rulebook agent that reads any plan you upload
A record that rest was given	Nothing we found	Confirmation from supervisor and workers

ShiftShield does not connect to those products. It brings their functions into one product and adds the two that are missing (shaded rows).

What changed across versions

Area	Version 1	Final plan	Why
Headline	Site WBGT alert	Site WBGT alert plus a two-sided rest record	The alert alone already exists abroad. The record does not.
Workers	Not in the loop	Anonymous confirmation by QR at the rest point	The named gap: no feedback loop from workers.
Rules	Compared with a city alert	Also checked against the 1 to 4 PM mandated window	Ties the product to a rule that is in force.
Output	Schedule and log	Schedule, rest record, heat-hours ledger	Gives officers and contractors a number to act on.
Backend	Python Lambdas plus Spring Boot on App Runner	Python only: one API Lambda behind API Gateway	One language, no idle cost.
Site setup	Nine-field form	Form first; photo suggestion if time allows	Photo setup films well but is no longer the centrepiece.
Shift re-planner	Not present	Kept from version 2 as P1	Gives the contractor a money reason to comply.
Site factors	Surface type weighted heavily	Sun and shade weighted most	A field study found on-site WBGT about 2.4 to 2.5 C above station estimates, with little difference between surface types.
Audit log	Hash-chained, P0	Append-only is P0; hash chain is P2	Invisible in a three-minute video.

Shaded rows are new since version 2.

The rule for this version

One loop, finished: alert, supervisor tap, worker confirmation, rest record. Plus one public page anyone can use. Everything else is built only if both are live by Saturday noon.

Executive summary

- **The problem.** Indian heat action plans issue one city-wide alert from dry-bulb air temperature at a single station. Heat injury is driven by wet-bulb globe temperature (WBGT), which also reflects humidity, radiant heat and air movement. Supervisors get a warning with no instructions.
- **The build.** ShiftShield estimates WBGT for one worksite from public forecasts and a site profile, converts it to a work-rest schedule from published guidance, alerts the supervisor about an hour before the schedule must tighten, and keeps a timestamped log.
- **What is new.** A rest record confirmed by the supervisor and, anonymously, by workers scanning a QR code at the rest point; an agent that turns the local heat action plan into a checkable rulebook, so the daily plan shows what the plan requires and what physiology adds, with the stricter of the two applied; and a ledger of work hours lost to heat.
- **Where the proof comes from.** A replay of a documented hot day for two site profiles against a dry-bulb baseline, a sensitivity analysis, and site parameters taken from cited literature.
- **Safety stance.** Decision support, not medical advice. A deterministic engine decides the schedule; the AI only reads photos, words messages and explains.

Honest expectation

No plan can guarantee a win: 4,034 people had registered when version 1 was written, and each track has one cash winner. This plan aims to put the entry in the top group and to give an interviewer something worth twenty minutes of questions.

SECTION 2

The hackathon: rules, judging and what recognition needs

From the event pages and rulebook. Re-read the Rules and Schedule pages on Thursday morning.

Item	What the organisers say	What it means for us
Format	Online Thu to Sun, Oct 8 to 11. Optional build day at DTU, Delhi, Sat Oct 10, limited seats.	Everything judged happens online. Delhi adds nothing to the score.
Eligibility	University students in India aged 18 or older, with a verified AWS Builder Center profile. Each teammate registers separately.	Finish student verification now. Rewards and the interview wait on it.
Teams	1 to 4 people. One submission per team.	Fill open seats before kickoff or commit to the two-person plan.
Originality	Project work starts when the clock does. A repo history that does not match the event dates disqualifies the team.	No repo or code before Oct 8. Research, design and account setup are allowed.
AWS	Use an AWS open-source tool or deploy on AWS. The video must show it.	Do both: Strands Agents SDK and SAM, plus a deployed stack shown in the console.
Submission	Public repo, demo video of up to 3 minutes, short writeup. Form opens Sun Oct 11, 8:00 AM.	Everything ready Saturday night. Submit soon after the form opens.

Judging criteria and how this plan answers them

Criterion	How ShiftShield answers it
1. Idea and impact	A gap no existing tool fills: a rest record confirmed by workers. Tied to a rule in force (the mandated afternoon break), backed by a backtest and by the words of a real supervisor and a real worker.
2. Built on AWS	EventBridge Scheduler, Lambda, API Gateway, DynamoDB, S3, SNS, Amplify Hosting and Bedrock, with Strands Agents SDK and SAM as open-source tools. Each has a one-line reason.
3. Design and usability	A message readable in ten seconds, one tap for the supervisor, one scan and one button for workers, all in the local language, with no login and no app to install.
4. Execution	One language, a deterministic core, replay mode, and a cutline agreed in advance.
5. Demo video	Two sites with two schedules, the alert, the supervisor's tap, workers scanning the QR, the rest record and ledger, then the AWS console. Script in Section 17.

Getting recognised beyond the track prize

- **Write the AWS Builder Center blog.** Five AirPods go to the best posts and fewer people will write one than submit a project. Keep build notes from Thursday so the post is quick to finish.
- **Be visible during the event.** Introduce the project in the WeMakeDevs space and Discord, and post progress each evening.
- **Fast-track interview.** At most ten students per hackathon, for Pre-Final Year (2028) and Final Year (2027), decided separately from prizes. Check that your batch qualifies and that registration details match the Builder Center profile.

What the pages say, and what it means for how we build

What the organisers say	What we do about it
"Judges score what you submit and nothing else. There is no live demo and no call."	The video, the README and the writeup are the product. They get a full half-day, not the last hour.
"One feature that runs beats five that almost do."	One loop end to end: alert, supervisor tap, worker confirmation, rest record. The re-planner and photo setup drop to optional.
"Would someone who isn't on your team know what to do with it?"	A public heat check on the live URL, no login, usable by a judge in ten seconds.
"Does it fix a real environmental problem? And what changes for the people living with it?"	A small field test with real workers and a supervisor, and their words in the video.
"A small problem solved well beats a big one solved vaguely."	One worksite, one shift, one city's rule. No multi-city or multi-site claims.
AWS use must be shown in the video; naming it in the writeup is not enough.	Twenty seconds of console: the schedule firing, the Lambda run, the DynamoDB items.
The judges for this event are not published yet. First Commit used 28 industry engineers, developer advocates and founders, two of them from Amazon. The mentors are AWS engineers.	Write for a mixed panel: a human story for the advocates and founders, sound architecture for the engineers.
For interviews, the project stands in for the screening round.	Clean commits from hour one, tests, a README a stranger can follow.

What the First Commit winners did (same tour, same rules, September 2026)

Prize	Project	What it does	What stood out in the repository
1st, Ship It	Suraksha (team of 3)	Voice companion for an ageing parent that alerts family only when needed.	14 AWS services, consent rules enforced with Cedar, an escalation chain in Step Functions, named limitations.
2nd, Build It	Beacon Night Shift (solo)	Voice agent that diagnoses a 3 AM outage and fixes it when the engineer says "approve fix one".	244 tests, 87 commits, CI, a demo that plays itself with no AWS account.
3rd, Best UI	CampusEvac (team of 3)	Two-player fire evacuation drill in the browser.	Polished interface, voice narration with Polly, emailed reports, 49 commits.
Runner-up	Factcade (solo)	Shows a small factory its losses in rupees when a machine fails and proposes a plan.	Deterministic maths, agent reads results, owner approves, rule-based fallback.

What we take from them

- **Safety and care themes won the top three places.** A heat-safety project is in the right territory.
- **The AI proposes, a person approves, code decides.** ShiftShield already works this way; say it in the README as a numbered safety model.
- **Voice.** Four of the seven prize winners use speech. The alert is spoken as well as written.
- **Escalation with timeouts.** The first-prize project escalates through a chain of people. We do the same when an alert is not acknowledged.
- **A human moment as the opening line.** "It is 1:40 PM on a concrete slab. The supervisor's phone says heatwave warning. It does not say what to do."
- **A demo that plays itself.** One link that runs a replayed hot day end to end, with no setup.
- **Money, not only minutes.** Work lost and saved is shown in rupees.
- **Visible engineering.** Winners had 35 to 87 commits, tests, infrastructure as code and AWS open-source tools doing real work.

No judges' comments are published, and the videos were not reviewed, so these are patterns read from the winners' repositories, not stated reasons.

The public door: a heat check anyone can use

- Open the URL, allow location or drop a pin, pick sun or shade and the kind of work. No login, no install.

- The page shows the site band for the next few hours with plain advice: how long to work, how long to rest, how much water, in English and one local language.
- It is the same engine and threshold file as the supervisor flow, so it costs one extra page, not a second product.
- It serves the people the problem statement names who have no supervisor: street vendors, delivery riders, loading crews.
- It is also what a judge will try first. Put the link at the top of the README and at the end of the video.

Engineering quality that AWS engineers will look for

- **One-command deploy:** the whole stack in a SAM template, with the deploy steps in the README.
- **Least privilege:** a separate IAM role per function; the log writer can append only.
- **DynamoDB design written down:** list the access patterns and the keys that serve them.
- **Failure handling:** idempotent writes, retries with a dead-letter queue, and a CloudWatch alarm on failed evaluations and stale forecasts.
- **Tests in CI:** engine, schedule and trigger tests run on every push with GitHub Actions.
- **A measured cost:** state the real cost per site per month from the billing console, not an estimate.

SECTION 3

The problem statement and the science behind it

Problem statement (as supplied by the Head of Department)

India's heat action plans issue one city-wide alert from a single weather station, based on dry-bulb air temperature. But heat injury is driven by wet-bulb globe temperature (WBGT), which combines humidity, radiant heat and air movement. A 38°C afternoon at 60% humidity on a reflective concrete site is physiologically far more dangerous than 42°C dry, and a city-level alert cannot tell those apart. The people who collapse are construction labour, street vendors, delivery riders and loading crews, working in micro-environments that run many degrees hotter than the station reading. Their supervisors get an SMS saying "heatwave warning" and have no idea what to actually do with the next 60 minutes.

Build a shift-level heat safety agent for a single worksite. It estimates site-specific WBGT from public weather data plus land-surface and land-use characteristics, converts it into an ILO/NIOSH-style work-rest schedule, pushes it to the supervisor's phone an hour before the threshold is crossed, and keeps a timestamped log of what was issued.

Phrase in the statement	What it requires	Section
site-specific WBGT	WBGT per time step from forecast temperature, humidity, wind and radiation, adjusted for the site.	5, 6
land-surface and land-use characteristics	Surface, shade and surroundings, now suggested from a photo and open land-cover data.	5
ILO/NIOSH-style work-rest schedule	A deterministic lookup from WBGT, work intensity and acclimatisation.	7
an hour before the threshold is crossed	A forecast-driven trigger with about 60 minutes of lead time.	9
timestamped log	An append-only record of inputs, values, message, delivery and both confirmations.	10, 13
agent	A tool-using AI component that words the plan, answers questions and suggests a profile from a photo, within safety limits.	11

Why WBGT, not air temperature

Outdoors, with sun : $WBGT = 0.7 * T_{nwb} + 0.2 * T_g + 0.1 * T_{db}$
 Indoors / no sun : $WBGT = 0.7 * T_{nwb} + 0.3 * T_g$
 T_{nwb} = natural wet-bulb, T_g = black globe, T_{db} = dry-bulb air temperature

Dry-bulb carries only 10 percent of the outdoor formula. Humidity drives the wet-bulb term, which carries 70 percent. A check of the statement's own example with Stull's wet-bulb approximation:

Condition	Dry-bulb	Humidity	Wet-bulb (approx.)	Reading
Humid afternoon	38°C	60%	about 31°C	Evaporation is hard; wet-bulb is high.
Dry heat	42°C	20%	about 24°C	Hotter air, but sweat evaporates easily.
Dry heat, more humid	42°C	30%	about 27°C	Still below the humid afternoon.

These are psychrometric wet-bulb values, not full WBGT. Reproduce the comparison with your chosen WBGT library and put the real numbers in the video.

Guidance to cite

Source	Use in this project
NIOSH, Criteria for a Recommended Standard: Occupational Exposure to Heat and Hot Environments (2016)	Primary transcription source. Free to read in full. Recommended limits by work rate and acclimatisation; work-rest and hydration practices. Note the page for every row you copy.
ILO, Working on a warmer planet (2019)	Problem framing: heat stress and work.
India's national and city heat action plan guidance (NDMA and city plans)	Describes the current alert, which is the baseline you improve on.
ACGIH TLVs for heat stress; ISO 7243	Background on work-rest allocations and the WBGT definition. Check what access the college library has before relying on either.

SECTION 4

Requirements, priorities and success metrics

P0 is the working product. P1 is what makes it stand out. P2 happens only if Saturday afternoon is calm.

ID	Requirement	Priority	Acceptance test
R1	Site profile: location, surface, shade, enclosure, work intensity, acclimatisation, shift hours, contact, language.	P0	Profile saved and used in the next evaluation.
R2	Site WBGT every 15 minutes from public forecast data, with an uncertainty margin.	P0	Series for the shift generated and displayed.
R3	Work-rest schedule from a versioned threshold file transcribed from NIOSH.	P0	Same inputs always give the same schedule; file version logged.
R4	Alert about 60 minutes before the schedule must tighten, with hysteresis and no duplicates.	P0	One alert per transition; lead time measured in replay.
R5	Email delivery to the supervisor.	P0	Message arrives once with correct numbers.
R6	Append-only, timestamped log of every issued plan.	P0	An entry exists for each message.
R7	Dashboard: WBGT chart with bands, current plan, log, comparison with the city-style alert.	P0	Supervisor and judges see the same numbers.
R8	Replay of a past hot day and backtest against a dry-bulb baseline.	P0	Chart and results table produced by a script.
R9	Rest record: supervisor one-tap acknowledgement plus anonymous worker confirmation by QR at the rest point.	P0	Each alert shows one of four states; no personal data stored.
R10	Symptom reports: workers tap dizzy, cramps or headache on the QR page; enough reports tighten the plan by one band.	P0	The plan can only get stricter; supervisor is alerted.
R11	Rulebook agent and daily compliance plan: obligations extracted from a heat action plan with source text and page, approved by a person, merged with the site WBGT (stricter wins), each marked met or not met.	P0 for two plans	Every approved rule shows its quote and page; a quote not found on the page is rejected.
R12	Public heat check and a self-playing demo link.	P0	A stranger gets advice in ten seconds; the demo link runs a full replay with no setup.
R13	Escalation chain: no acknowledgement in 15 minutes goes to the safety officer, then the contractor.	P1	Chain runs in replay with shortened timers.
R14	Voice alert in the local language.	P1	Audio plays from the alert; text is always sent as well.
R15	Heat-day certificate: signed daily record for the site, checkable by QR.	P1	Changing any figure makes verification fail.
R16	Shade payback and the heat-hours ledger in rupees.	P1	Figures match the engine output.
R17	Agent wording in two languages and 'why' answers.	P1	Numeric consistency check passes; template fallback works.

ID	Requirement	Priority	Acceptance test
R18	WhatsApp API, shift re-planner, photo-to-profile, hash chain, sensor check, missed-call confirmation.	P2	Mention as next steps.

Replay moved up to P0 because October weather will not reach the thresholds: without replay there is no demo.

Success metrics (report only what you measure)

Metric	Definition	Intent
Missed-danger hours	Hours where the site band is High or above while the dry-bulb baseline is silent.	Show what the city alert misses.
Needless-alarm hours	Hours where the baseline alarms while the site band is Low.	Show what the city alert overstates.
Lead time	Minutes between the alert and the forecast schedule change, from replay.	About 60 minutes.
Work minutes kept	Safe work minutes in the re-planned shift minus those in the original order.	The contractor's reason to comply.
Confirmation rate	Alerts with a supervisor tap, and alerts with worker confirmations, divided by alerts sent, in testing.	Evidence the loop closes.
Danger outside the window	Hours at High or above that fall outside 1 to 4 PM, from the replay.	Show what a fixed rule misses.
Correctness	Duplicate alerts, and messages whose numbers differ from the engine output.	Zero in tests.
Sensitivity	How often the band changes when site assumptions vary within a stated range.	Show the system is robust, or cautious where it is not.

Definition of done

A stranger opens the live URL and gets heat advice for their own spot without logging in. A supervisor sets up a site, sees the forecast and schedule, receives one correct alert and taps to acknowledge; a second phone scans the rest-point QR and confirms the break; an officer reads the rest record and the day's ledger. All on AWS, in a video under three minutes.

SECTION 5 | ADDED IN V2

Site profile and photo-to-profile setup

How a photo and a 30-second confirmation turn a city forecast into a site estimate.



How it works

- The supervisor takes one or two photos of the work area. The browser uploads them to S3 through a presigned URL.
- A Lambda sends the image to a multimodal model on Amazon Bedrock with a constrained prompt. The model must return only a small JSON object: ground surface, shade level, roof or enclosure, and a confidence for each. Confirm which image-capable models your account can use in your region before kickoff.
- The suggestion appears as a pre-filled profile. The supervisor confirms or changes each field. Nothing is saved without that confirmation.
- Low confidence, an unreadable image or a failed call falls back to the cautious default for that field and the manual form.

Profile field	Options	From photo?	Default if unknown
Location	Map pin or coordinates	No, from the phone or a pin	Required
Ground surface	Bare soil, grass, asphalt, light concrete, dark concrete, mixed	Yes	Concrete (more cautious)

Profile field	Options	From photo?	Default if unknown
Shade	None, partial, mostly shaded, indoor	Yes	None
Roof or enclosure	Open, metal roof, concrete roof, ventilated, closed	Yes	Open
Work intensity	Light, moderate, heavy	No, supervisor chooses	Heavy for construction and loading
Acclimatisation	Mostly acclimatised, or new or returning workers present	No	New workers present
Clothing or PPE	Normal, heavy or impermeable	No	Normal
Shift and tasks	Start, end, and a short task list with intensity and duration	No	Required; feeds the re-planner
Contact and language	Email, WhatsApp number, English or local language	No	Email and English

Guardrails for the photo step

- The photo only suggests profile fields. It never touches thresholds, schedules or recipients.
- Ask for a photo of the site, not of people. Store the labels and delete the image after the profile is confirmed, so no worker images are kept.
- The model's output is validated against the allowed options. Anything else is rejected and the default is used.
- The log records that a field came from a photo suggestion and whether the supervisor changed it.

Site adjustment model

```

S_site = (1 - shade)*S_direct + S_diffuse + k_surface*S_total # W/m2 at the globe
T_site = T_air + dT_landuse # built-up or bare surroundings run warmer
RH_site = rh_from(T_site, e(T_air, RH)) # same moisture, warmer air, lower RH
u_site = f_wind * u_10m # sheltered sites get less wind
WBGT_site = wbgt_method(T_site, RH_site, u_site, S_site)

```

Parameter	Starting range (assumptions to replace with cited values)	Direction
shade	0 (none) to about 0.8 (mostly shaded)	More shade lowers the radiant load.
k_surface	Small for grass (about 0.05), larger for light concrete (about 0.20 to 0.25)	Reflective surfaces add reflected and re-radiated heat.
dT_landuse	About 0 (vegetated) to +2°C (dense built-up)	Dense built-up raises local air temperature.
f_wind	About 1.0 (open) down to 0.4 (enclosed or sheltered)	Less wind means less cooling.

Evidence to respect: a field study at Arizona State University measured on-site WBGT about 2.4 to 2.5°C above estimates from regional weather stations, and found the differences between five surface types largely insignificant. So weight sun and shade most, keep the surface factor small, and say so.

Do this before Thursday

These ranges are placeholders, and a judge can dismiss the whole idea if they stay that way. Finding published values is allowed pre-kickoff research: surface albedo tables for k_surface, urban heat island studies for Indian cities for dT_landuse, and shade and wind reduction factors from WBGT or urban-climate papers. Keep one source and page number per parameter in a notes file, ready to commit on Thursday.

SECTION 6

Weather data and the WBGT engine

Established methods first. The team's contribution is the site adjustment, the uncertainty handling and the testing.

Need	Source	Notes
Hourly forecast: temperature, humidity, wind, shortwave, direct and diffuse radiation	Open-Meteo forecast API	Free for non-commercial use with attribution. Resolution is coarse; say so.
Historical hot day for replay and backtest	Open-Meteo historical archive	Reanalysis-based; may differ from station readings.
Site surroundings (P2)	ESA WorldCover on the AWS Open Data registry, OpenStreetMap	Optional built-up and vegetation fractions. Verify the path and licence.

Method choices

Option	Description	Use
A. Physical outdoor model	A published model for natural wet-bulb and globe temperature from temperature, humidity, wind and radiation (the Liljegren approach is widely used). Libraries such as thermofeel and pythermalcomfort provide WBGT functions; confirm function names, inputs and licences on Thursday.	Primary.
B. Simple approximation	A temperature-and-humidity formula with no radiation or wind term.	Cross-check only.
C. Dry-bulb rule	A threshold on air temperature modelled on current heat-action-plan practice.	Baseline for the backtest.

Engine steps (pure functions, same code for live and replay)

- Fetch the hourly series for the shift window and interpolate to 15-minute steps.
- Apply the site adjustment from Section 5 to radiation, air temperature, humidity and wind.
- Compute WBGT with method A. Record the method and library version.
- Attach a margin: $WBGT_{conservative} = WBGT + \text{margin}$. Start near 1°C and justify the final value with the sensitivity analysis.
- Output per time step: WBGT, margin, inputs used, data age and method version.

Tests that earn credibility

- **Reference cases:** compare against published example values for the chosen model, within a stated tolerance.
- **Monotonicity:** more humidity must not lower WBGT; more shade must not raise it; stronger wind must not raise it.
- **Regression:** freeze a set of inputs and outputs so later changes cannot silently alter issued schedules.

SECTION 7

The work-rest schedule engine

The safety-critical part: deterministic, cited, versioned, and cautious when uncertain.

The engine takes the conservative WBGT, the work intensity, the acclimatisation status and any clothing adjustment, and returns a band with a full instruction. The mapping lives in a data file (for example thresholds.v1.json) transcribed from NIOSH 2016, with the page noted for every row.

Band	Meaning	Work and rest per hour	Water	Extra controls
A Normal	Below the lowest limit for this work and acclimatisation.	Normal work with standard breaks.	Drink regularly.	Awareness reminder.
B Caution	Approaching the limit.	[from the table]	[from the table]	Shade rest area, buddy check at breaks.
C High	At or above the limit.	[from the table]	[from the table]	Move heavy tasks to cooler hours.
D Very high	Well above the limit.	[from the table]	[from the table]	Stop non-essential heavy work.

Square brackets mean the value must come from the cited guidance. The layout is the design; the numbers are not the team's to invent.

- **Cautious at boundaries:** ties go to the stricter band, and the lookup uses the conservative WBGT.
- **No silent relaxing:** an easier schedule is issued only after hysteresis confirms it, and is labelled as an easing.
- **Fail-safe on bad data:** a stale forecast produces a message saying so and the most cautious plan consistent with the last good forecast.
- **Fixed safety footer:** every message ends with the same first-aid line, appended by code: shade, cool the person, water if conscious, call 112 for confusion, fainting or hot dry skin.
- **Reviewed:** ask an occupational health or safety professional, through the HOD if possible, to check the file before the demo.

SECTION 8 | ADDED IN V2

The shift re-planner

The same hours, re-ordered, so safety costs less work. Pure calculation on data the engine already has.

A work-rest schedule tells the supervisor how much work is lost. The re-planner shows how to lose less of it. It takes the task list from the site profile and the WBGT series for the day, and proposes an order and a start time that keep the most safe work minutes.

Inputs and outputs

Item	Detail
Task list	Three to six tasks for the shift, each with an intensity (light, moderate, heavy) and a duration. Entered once at setup.
Allowed changes	Supervisor ticks what is possible: start up to two hours earlier, move the long break, re-order tasks.
WBGT series	The conservative 15-minute series from the engine for the whole day.
Output	Original plan and re-plan side by side: order of tasks, work and rest minutes per hour, and total safe work minutes for each.

Method (deterministic)

- For any hour, the safe work minutes for a task are read from the threshold file using that hour's band and the task's intensity.
- Rank the hours of the candidate shift window from coolest to hottest by conservative WBGT.
- Assign heavy tasks to the coolest hours, moderate tasks next, light tasks to the hottest hours. Place the long break in the hottest hour.
- Repeat for each allowed start time and keep the option with the most safe work minutes. Ties go to the option with less time in Band C or D.
- Report the difference. If the re-plan gains fewer than a set number of minutes, say the current order is already fine.

```
# pseudocode (design only)
def safe_minutes(order, start, series):
    return sum(work_minutes(band(series, hour), task.intensity)
               for hour, task in place(order, start))

best = max(candidates(tasks, allowed_starts),
           key=lambda c: safe_minutes(c.order, c.start, series))
```

What the supervisor sees (illustrative numbers)	Original order	Re-plan
Shift	8:00 to 17:00	6:30 to 15:30
Slab casting (heavy)	11:00 to 14:00, Band C and D	6:30 to 9:30, Band A and B
Material shifting (moderate)	8:00 to 11:00	9:30 to 12:30
Marking and checks (light)	14:00 to 17:00	12:30 to 15:30, with the long break at 13:00
Safe work minutes	X	Y, which is Y minus X more

X and Y come from the replay, not from this document. The headline for the video is the measured difference.

Guardrails

- The re-planner only re-orders. It never loosens a work-rest ratio; every hour still follows the threshold file.
- It is advice. The supervisor decides, and a re-plan that is not accepted changes nothing in the alerts.
- Property test: for any input, the re-plan has at least as many safe work minutes as the original order, and no hour exceeds its band limit.
- The agent may explain the re-plan in words. It does not compute it.

SECTION 9

Alert logic and the one-tap acknowledgement

Alert about an hour before the schedule must tighten, once, then confirm someone acted on it.

Message type	When it fires	Notes
Shift plan	About 30 minutes before shift start.	Hour-by-hour plan and the re-plan, if it helps.
Heads-up	The next band change is forecast about 60 minutes away (window 45 to 75 minutes).	States the time, the new schedule and the reason. Carries the acknowledge link.
Easing	Conditions have improved for two consecutive evaluations, below the threshold minus a buffer.	Labelled as an easing; never sent on a single reading.
Data issue	Forecast stale or unavailable beyond a set time.	Recommends the cautious plan; says what is missing.

Rules that keep the system quiet and correct

- **Re-evaluate every 15 minutes.** If the predicted crossing moves, the pending heads-up moves with it; if it comes within 30 minutes, send immediately.
- **Hysteresis:** tightening uses the threshold; easing needs a buffer below it for two consecutive evaluations.
- **Cooldown:** no repeat of the same message type within 30 minutes. Tightening to a higher band always bypasses it.
- **Idempotency:** key = siteId + shiftDate + fromBand + toBand + windowStart, claimed with a DynamoDB conditional write, so a Lambda retry cannot double-send.

```
# pseudocode (design only)
band_now, band_next, t_change = next_band_change(series, profile)
if band_next > band_now and 45 <= minutes_until(t_change) <= 75:
    key = f'{site.id}#{date}#{band_now}#{band_next}#{t_change:%H%M}'
    if put_if_absent(alert_keys, key):
        issue(site, band_next, t_change, series) # word it, deliver, log
```

One-tap acknowledgement (new)

Every heads-up carries a link with a signed, single-use token. Tapping it opens a page with one large button, "Break started", and records the time against that alert. No login.

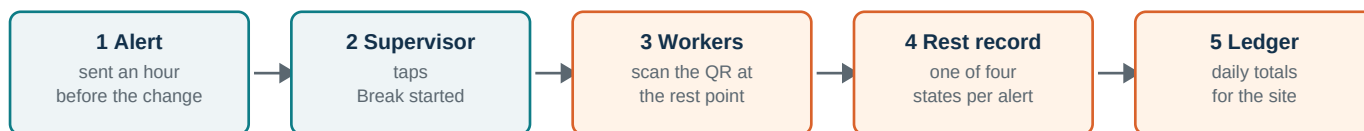
State	Meaning	What happens
Sent	Alert issued and delivered.	Dashboard shows it as waiting.
Acknowledged	Supervisor tapped the link.	Time recorded as a new log entry linked to the alert.
Not acknowledged	No tap within 20 minutes of the schedule change.	One reminder is sent. After that it stays marked as unconfirmed; the system does not nag.

- The token is tied to one alert and one site, expires at shift end, and can only add an acknowledgement. It cannot change a profile, a threshold or a recipient.
- The acknowledgement records what the supervisor says happened. It collects no worker-level data and is not used to rate anyone.
- If WhatsApp is live, a reply of "1" can do the same job. That depends on inbound message handling in AWS End User Messaging Social, so treat it as optional and verify how inbound events are delivered.

SECTION 10 | NEW IN V3

The rest record, the rule check and the heat-hours ledger

The centrepiece. The alert says what should happen; this records what did.



How worker confirmation works

- A printed QR code is fixed at the shaded rest area. It encodes a site code, not a person.
- Any phone can scan it, including one shared phone. The page opens in the local language with two large buttons: "We got the break" and "No break". Two optional yes or no questions follow: is there water, is there shade.
- No login, no name, no phone number. A random token in the browser limits each phone to one response per break window.
- The page accepts responses only during a break window the engine has issued, so old or early scans count for nothing.
- Only totals are stored and shown, and the yes or no split is hidden until at least three responses arrive.

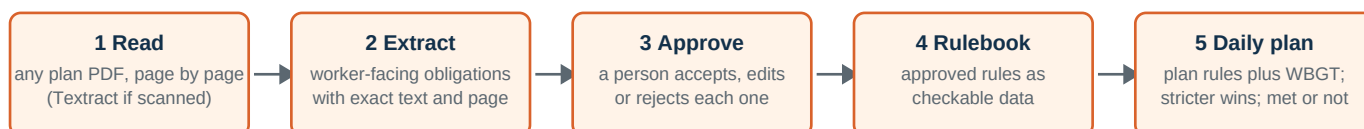
State of an alert	Meaning	Shown as
Confirmed by both	Supervisor tapped and most worker responses say the break was given.	Green tick with the count.
Supervisor only	Supervisor tapped; no worker responses.	Amber: unverified.
Disputed	Supervisor tapped, but most worker responses say no break.	Red flag for the safety officer.
No record	Neither side responded.	Grey: unconfirmed.

Honest limits and guardrails

- It is evidence, not proof. A supervisor could scan the code themselves; the time window, the per-phone limit and the three-response minimum make that harder, not impossible. Say this in the video.
- Workers must not be exposed. No identity is collected, individual responses are never shown, and the record is not used to rate workers.
- Workers without phones: one shared phone at the rest point is enough. A missed-call number is a stretch item only.
- A browser location check (is the phone near the site) is optional and off by default, because it asks for a permission that may put workers off.
- Talk to two or three workers before Thursday. If they would not scan it, change the design before you build it.

Heat action plan rulebook agent and the daily compliance plan

Heat action plans are long and differ by city and state, so a supervisor cannot tell which obligations apply to the site today. The agent reads a plan and turns it into rules the system can check. This replaces the fixed 1 to 4 PM setting of earlier versions.



- **Extract.** For each obligation the agent returns who it binds, what must be done, when it applies (hours, alert level or temperature), the exact sentence and the page number.
- **Verify the quote in code.** The quoted sentence must be found on the stated page of the document text. If it is not, the obligation is rejected before a person ever sees it. The model cannot invent a rule.
- **Approve.** A person (safety officer or team member) accepts, edits or rejects each obligation. Only approved ones enter the rulebook, with who approved and when.
- **Rulebook.** Each rule is data: a condition, a requirement and a way to check it. Example fields: `applies_when`, `requirement_type` (rest window, water, shade, working hours, first aid), `parameters`, `source quote`, `page`, `plan version`.

- **Daily compliance plan.** For today's forecast the engine lists what the plan requires and what the WBGT schedule adds, and applies the stricter of the two for every hour.
- **Record.** Each obligation is marked met, not met or unconfirmed, with a timestamp. Rest comes from the two-sided rest record; water and shade from the workers' answers on the QR page; the rest from a supervisor checklist.

Obligation (illustrative)	Plan says	Physiology says	Applied today	Confirmed by
Afternoon rest	No outdoor work 1 PM to 4 PM	High band from 11:30 AM	Work-rest cycles from 11:30; full stop 1 to 4	Supervisor tap and worker QR
Drinking water	Water at the worksite	A cup every 15 to 20 minutes at this band	Water point plus timed reminders	Workers' answer on the QR page
Shade	Shaded rest area	Rest must be out of direct sun	Shaded rest area	Workers' answer on the QR page

Illustrative only. Real rows come from the approved rulebook and the day's WBGT series.

Scope and guardrails

- The agent accepts any heat action plan you upload. Two plans (Delhi's 2026 plan and your own state's or city's) are checked against a hand-made answer key, so say "works on any plan, verified on two".
- Before Thursday, read both plans and list the worker-facing obligations by hand with page numbers. That list is research, and it becomes the test the agent is scored against.
- The law comes first. ShiftShield never advises work during a mandated rest period; physiology can only add protection.
- The rulebook is a reading aid, not legal advice. Every rule shows its source sentence and page so a person can check it.
- Fallback: if extraction is unreliable on demo day, ship the two approved rulebooks as data and show extraction on one short section.

Heat-hours ledger

Column (per site, per day)	Source
Rest minutes prescribed	Schedule engine, summed over the shift.
Rest minutes confirmed	Alerts in the state 'confirmed by both'.
Work minutes lost to heat	Shift length minus safe work minutes.
Danger hours outside the mandated window	Rule check.
Water and shade reported available	Share of worker responses saying yes.

The ledger is a query over the log, exported as CSV. It is the figure a contractor, a labour officer or an insurer would ask for. ShiftShield does not pay anyone or decide compensation; it supplies the record.

Symptom reports tighten the plan

- The worker page adds three buttons: dizzy, cramps, headache. They are optional and anonymous, like the rest confirmation.
- If a set number of phones report a symptom in one break window (start with three), the site moves up one band for the rest of the shift and the supervisor is alerted.
- **Tighten only.** Reports can make the plan stricter and can never relax it. A lack of reports means nothing.
- This is the answer to "how do you know your estimate is right without a sensor?": the people on site correct it.
- It is not diagnosis. The page shows the fixed first-aid steps and the 112 call button to anyone who taps a symptom.

Escalation chain

- An alert that is not acknowledged within 15 minutes of the schedule change goes to the safety officer; after another 15 minutes, to the contractor.
- Built as an AWS Step Functions workflow that waits for the acknowledgement token and times out to the next person.
- Each step is written to the log. In replay the timers are shortened so the chain can be shown in seconds.

Heat-day certificate

- At the end of the shift the site gets a one-page record: hours at High or above, rest prescribed, rest confirmed by both sides, work minutes lost and their value in rupees.
- The record is signed with a key held in AWS KMS. A QR code on the page opens a verify screen that recomputes the check, so an edited certificate fails.
- The one heat insurance scheme we found pays on city air temperature. This is site-level evidence a welfare board or insurer could use. ShiftShield issues the record; it does not pay or decide anything.

Shade payback

- Re-run the engine for the same day with the shade setting switched on, and show the difference: "A shade net over this area gives back X work minutes a day, worth about Y rupees."
- Rupees are crew size times daily wage times minutes kept, divided by shift minutes. The supervisor enters crew size and wage once.
- X and Y come from the engine, never from this document.

SECTION 11

The agent: three jobs, a short leash

It reads the photo, words the plan and explains. It never decides safety.

Decision	Made by	Why
Whether to alert now	Deterministic trigger	Testable, repeatable, auditable.
Which schedule applies	Deterministic schedule engine	Comes from the cited table; no model discretion.
Which task order is best	Deterministic re-planner	A calculation, checked by a property test.
What the site looks like	Agent suggests, supervisor confirms	Vision is where a model saves the user time.
Which obligations a heat action plan contains	Agent extracts, code verifies the quote, a person approves	Reading long documents is the agent's essential job here.
How to say it, in which language	Agent, with a template fallback	Language is where a model helps.
Explaining 'why' and 'what if'	Agent calling tools	Answers are grounded in tool outputs.

Tools the agent can call

Tool	Purpose
<code>suggest_profile(photo_ref)</code>	Return surface, shade and enclosure labels with confidence, from the allowed options only.
<code>get_wbgt_series(site_id, from, to)</code>	Run the site WBGT engine for a time window.
<code>extract_obligations(plan_id, pages)</code>	Return candidate obligations with exact quote and page; quotes are verified in code before review.
<code>daily_compliance_plan(site_id, date)</code>	Merge the approved rulebook with the WBGT schedule; the stricter requirement wins.
<code>lookup_schedule(wbgt, intensity, acclimatisation, clothing)</code>	Query the threshold file. Returns band, minutes, water interval, controls and the citation.
<code>replan_shift(site_id, allowed_changes)</code>	Run the re-planner and return both plans with their totals.
<code>render_message(plan, language)</code>	Fill the approved template. Used as the fallback.

Guardrails

- **Numeric consistency check:** every number and time in a generated message is parsed and compared with the plan. Any mismatch rejects the message and the template is sent instead.
- **Constrained prompt:** use only facts from the plan object, stay under about 50 words, make no medical claims.
- **Allow-listed recipients and tools:** the agent cannot message anyone unregistered, and cannot edit the threshold file or the log.
- **Untrusted input:** photos and any inbound message are data, not instructions.

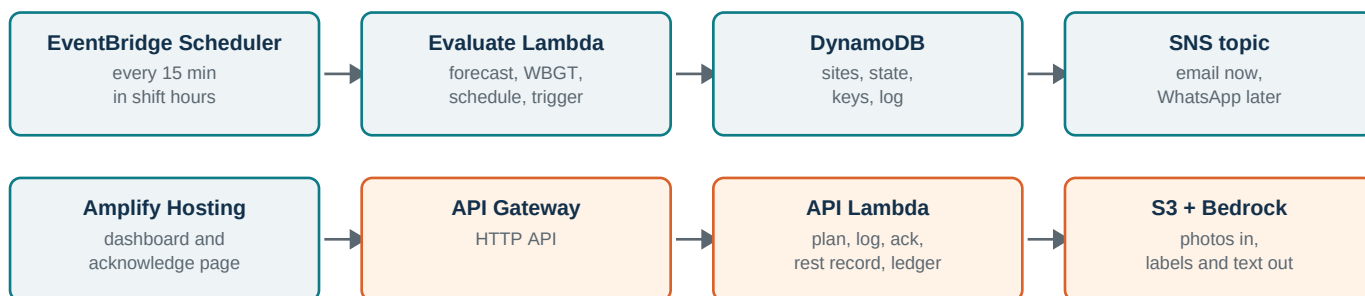
- **Graceful degradation:** if Bedrock is slow, unavailable or not yet approved, the template path and the manual form produce the same result, so the system never depends on the model.
- **Language review:** have a fluent speaker check the local-language templates before the demo.

Built with the Strands Agents SDK and a model on Amazon Bedrock. Request model access now, because approval can lag. About 20 golden scenarios are checked by script in both languages.

SECTION 12

AWS architecture, trimmed

Serverless throughout. One language, one deploy command, nothing that runs when idle.



Service	Role	Why this choice
EventBridge Scheduler	15-minute trigger during shift windows; replay payload for demos.	No servers; easy to pause.
Lambda (Python)	Evaluate function and API function.	The WBGT libraries are Python. Check library packaging in a layer early on Thursday.
API Gateway (HTTP API)	Routes for the dashboard, re-planner and acknowledge link.	Replaces the App Runner service; no idle cost.
DynamoDB	sites, site_state, alert_keys, issue_log, acks.	Conditional writes give idempotency.
S3	Site photos (short-lived), replay inputs, backtest outputs.	Presigned upload from the phone.
Bedrock + Strands Agents SDK	Photo labels, message wording, explanations.	A real AI feature and an AWS open-source tool.
SNS	Email delivery to supervisor, safety officer and contractor.	Decouples decision from delivery.
Step Functions	Escalation chain that waits for an acknowledgement and times out to the next person.	The pattern the first-prize project used.
Polly	Spoken version of the alert.	Check which Indian languages its voices cover before promising one.
KMS	Signs the heat-day certificate.	Verification without sharing a secret.
Textract	Reads scanned pages of a heat action plan.	Needed only when the PDF has no text layer; check language support for non-English plans.
Powertools for AWS Lambda	Structured logs, metrics and idempotency helpers.	AWS open-source, used by the Build It winner.
Amplify Hosting	Dashboard, acknowledge page and the worker page behind the QR.	Deploys from the repo with HTTPS.
CloudWatch, Secrets Manager, SAM	Alarms and logs; credentials; infrastructure as code.	SAM also counts as an AWS open-source tool.
End User Messaging Social (P1)	WhatsApp through a linked WhatsApp Business Account.	SNS cannot send WhatsApp itself.
IoT Core (P2)	Sensor readings.	Only if the sensor check is built.

Removed from version 1

Spring Boot on App Runner, ECR and the Docker build. If the team is far stronger in Java than Python, keep the API in Java as a single Lambda instead, but do not run a second always-on service.

SECTION 13**Data model, API and the log**

Table	Keys	Main attributes
sites	PK siteId	Location, profile, which fields came from a photo, task list, shift hours, language, contact (hashed), profileVersion.
site_state	PK siteId	currentBand, since, lastAlertAt, lastEvaluationAt, forecastAge. Source of truth for hysteresis.
alert_keys	PK key	createdAt, ttl. Conditional put for idempotency.
issue_log	PK siteId, SK ts#logId	Type, inputs reference, WBGT summary, margin, band, schedule, thresholdsVersion, engineVersion, message text, language. Append-only: the writer role cannot update or delete.
acks	PK logId, SK ts	Acknowledged time, channel, token id. Kept separate so the log stays immutable.
rest_confirms	PK logId, SK window	Counts only: yes, no, water yes, shade yes. Updated with atomic counters. No device or person identifiers stored.
rulebooks	PK planId, SK ruleId	Requirement type, condition, parameters, source quote, page, plan version, status (candidate, approved, rejected), approver and time.
obligation_log	PK siteId, SK date#ruleId	Applied requirement, source (plan, physiology or both), status (met, not met, unconfirmed), evidence reference, timestamp.
replay_runs	PK runId, SK ts	Scenario, site profile, outputs. Feeds the backtest and the demo.

API routes (one Lambda behind API Gateway)

Route	Purpose
POST /sites, PUT /sites/{id}	Create or update a site profile with validation.
POST /sites/{id}/photo	Return a presigned upload URL, then the suggested profile.
GET /sites/{id}/forecast	WBGT series with margin, bands and data age.
GET /sites/{id}/plan	Current and next schedule with reasons and the citation.
POST /sites/{id}/replan	Original plan and re-plan with totals.
GET /sites/{id}/log	Paged log entries with delivery and acknowledgement status.
GET /sites/{id}/compare	Site band versus the dry-bulb baseline for the same hours.
POST /ack/{token}	Record the supervisor's acknowledgement. Single use.
GET /rest/{siteCode}, POST /rest/{siteCode}	Worker page behind the QR code; record an anonymous response during an open break window.
POST /plans, GET /plans/{id}/candidates	Upload a heat action plan; list extracted obligations with quote and page.
POST /plans/{id}/rules/{ruleId}	Approve, edit or reject one obligation.
GET /sites/{id}/compliance	Daily compliance plan and the status of each obligation.
GET /sites/{id}/ledger	Daily heat-hours ledger, with CSV export.
GET /cert/{id}, GET /cert/{id}/verify	Heat-day certificate and its signature check.
GET /sites/{id}/shade	Shade payback for a day.
GET /demo	Self-playing replay of a hot day.
POST /replay/start	Run a replay scenario for the demo and backtest.

Practices: input validation on every write, CORS limited to the frontend, a simple token on officer routes, and contact details never returned by any GET. Log in UTC with IST beside it.

Hash chain, now P2

Each entry would store hash = SHA256(previous hash + canonical JSON of the entry), with a Verify button that recomputes the chain. It makes edits detectable and is good interview material, but it adds nothing a judge can see in three minutes. Build it only after the freeze items are done, and describe it as planned work if it is not built.

SECTION 14

Validation: replay, backtest, sensitivity and a sensor check

Without ground truth, the honest evidence is relative: this site against that one, and both against the city alert.

- **Replay.** Feed hourly historical weather for a documented hot-season day (April or May, in your pilot city) through the same engine, trigger, agent and log. A clear Replay banner shows nothing is passed off as live.
- **Two profiles, same weather:** a reflective open concrete site with no shade, and a shaded or vegetated site.
- **Baseline:** a dry-bulb rule modelled on current heat-action-plan practice. Cite the rule you copy.
- **Control day:** a milder day, to check the system stays quiet.
- **Sensitivity:** vary shade, surface factor, land-use offset, wind factor and margin across their cited ranges, one at a time and then together. Report the share of the shift where the band is stable.

Measure (fill in from the replay)	Baseline	Concrete site	Shaded site
Hours with site band High or above	n/a	X	X
Missed-danger hours	-	X	X
First heads-up and lead time	one daily alert	X min	X min
Rest minutes prescribed over the shift	none	X	X
Danger hours outside the 1 to 4 PM window	n/a	X	X
Safe work minutes: original order versus re-plan	n/a	X versus Y	X versus Y
Control day: alerts issued	X	X	X

The rest record cannot be backtested. Test it live: team members and a few volunteers scan the QR during a replayed break window, and report the confirmation counts.

Optional sensor check (only with hardware experience on the team)

- A microcontroller with a temperature and humidity sensor, plus a temperature probe inside a matt-black hollow ball, gives a rough globe temperature. Readings go to AWS IoT Core.
- Log a few hours in full sun and in shade on campus. Compare the direction and size of the sun-to-shade difference with what the engine predicts for the same hours.
- A home-made globe is not a calibrated instrument. Present it as a sanity check on the site adjustment, not as a WBGT measurement.
- Buying parts and learning the wiring before Thursday is fine. Firmware and cloud code for the project start with the clock.

SECTION 15

Safety, privacy, reliability and cost

Area	Practice
Safety positioning	Decision support, not medical advice or a replacement for official guidance. Say so in the product, the video and the writeup. The supervisor can always choose a stricter plan.
Conservative by default	Unknown inputs default to the cautious value; margin applied; ties go to the stricter band; no silent relaxing.
Photos	Photograph the site, not people. Keep the labels, delete the image after confirmation. Say this on the upload screen.
Privacy	Collect only the supervisor's contact and the site profile. No worker-level data. Explicit opt-in with timestamp, easy opt-out and deletion.
Rest record	Worker responses are anonymous counts tied to an alert. No names, numbers or device identifiers; the split is hidden below three responses; never used to score or discipline anyone.
Secrets and IAM	No secrets in the public repository; credentials in Secrets Manager; least-privilege roles; the log writer can append only.
Reliability	Idempotent writes, retries with a dead-letter queue, alarms on evaluation failures and forecast age, email baseline, replay for demo day.
Language and access	Local language, plain words, large text. Colour is never the only signal. A voice version through Amazon Polly is a stretch item; check which Indian languages its voices cover before promising one.

Cost control

- A new AWS account gets up to \$200 in free credits, which the organisers say counts in full. Set an AWS Budgets alert on day one.
- With App Runner removed, nothing in the stack runs when idle. Bedrock is the main variable cost: one image call per site setup and a few short messages per shift.
- WhatsApp is billed through AWS and Meta; check both before sending beyond team test numbers.
- Keep the deployment live until judging is finished.

SECTION 16

Four-day schedule, team split and cutline

Submission opens Sunday Oct 11 at 8:00 AM. Adjust when the schedule page publishes kickoff and deadline hours.

Day	Goals	Exit criteria
Thu Oct 8	Create the public repo and README; SAM skeleton; Budgets alert. Forecast fetch, WBGT engine with tests, threshold file with page references, cited site parameters. First WBGT chart for two profiles. Confirm the WBGT library packages into a Lambda layer. Commit the hand-made obligation list for two plans as test data.	Engine passes tests; chart renders locally; first commits in history.
Fri Oct 9	Deploy the evaluate Lambda, DynamoDB and the schedule. Trigger with hysteresis and idempotency, append-only log, email alert with the acknowledge link. API Lambda and routes. Dashboard skeleton and the public heat check on Amplify. Replay mode. Rulebook agent: extract, verify quotes, approval screen, daily compliance plan.	A scheduled run on AWS; one correct email alert; replay produces a day of bands.
Sat Oct 10	Morning: worker QR page with symptom buttons, rest record states, dashboard panel, self-playing demo link. Midday: field test with real workers, backtest and sensitivity run. Afternoon, in this order and only while time allows: escalation chain, voice alert, heat-day certificate, shade payback, agent wording. README and alarms. Freeze at about 9 PM.	End-to-end flow live on the URL; backtest numbers in hand.
Sun Oct 11	Record the video, upload, test the link signed out, finish README, credits and writeup. Submit soon after the form opens. Publish the Builder Center blog.	Submission confirmed.

Who owns what

Team size	Person	Owns
Two	A: engine and data	WBGT engine, site adjustment, threshold file, rule check, ledger, re-planner, replay, backtest, video and writeup.
	B: cloud and product	SAM stack, Lambdas, trigger, log, API, dashboard, acknowledge link, worker QR page and rest record, agent.
Four	C: frontend and agent	Takes the dashboard, worker QR page, photo flow and agent from B.
	D: evidence and story	Takes backtest, sensitivity, supervisor interview, video, writeup and blog from A. Owns the sensor check if anyone does.

Cutline: drop in this order

Order	Item	Fallback
1	WhatsApp API, re-planner, photo setup, hash chain, sensor check	Mention as next steps. A plain share link covers WhatsApp.
2	Agent-written wording	Template messages, which must exist anyway.
3	Shade payback, ledger export	Show totals on the dashboard.
4	Heat-day certificate	Show the daily record unsigned.
5	Voice alert	Text alert only.
6	Escalation chain	One reminder to the supervisor.
Never	Engine, schedule, trigger, email alert, log, dashboard, replay and demo link, rulebook and daily compliance plan for two plans, rest record with symptom buttons, public heat check, video	If these are not done by Saturday 2 PM, stop everything else.

Checkpoints: Fri 6 PM (scheduled run, alert and log work), Sat 2 PM (dashboard on the live API, rest record working end to end), Sat 9 PM (freeze). Commit continuously from kickoff and do not rewrite history.

SECTION 17

The three-minute video

There is no live demo, so this is what the judges see. Hard limit under 3:00; YouTube public or unlisted.

Time	On screen	Narration
0:00 to 0:20	A hot worksite (own or credited footage). The heatwave SMS. Title.	"It is 1:40 PM on a concrete slab. The supervisor's phone says heatwave warning. It does not say what to do." One line from the supervisor you interviewed.
0:20 to 0:50	Two sites, one hour: WBGT chart and two different schedules, with the city alert beside them.	How site WBGT works and why one site needs more rest.
0:50 to 1:10	Replay: the heads-up arrives 60 minutes early. Supervisor taps "Break started".	What the supervisor does with it.
1:10 to 1:45	A second phone scans the QR at the rest point and taps "We got the break". The alert turns to 'confirmed by both'. Then show a disputed one.	"Others send the warning. We record whether the rest was given, and the workers confirm it." No names, only counts.
1:45 to 2:05	The agent reads a heat action plan: an obligation appears with its sentence and page, a person approves it. Then the daily compliance plan: plan rule, physiology, stricter applied, met or not.	"No one could tell a supervisor which rules apply today. Now the plan is a checklist, and physiology adds what it misses."
2:05 to 2:40	AWS console: schedule, Lambda run, DynamoDB items, Bedrock call. Architecture diagram. Backtest table.	Where AWS fits, one line per service.
2:40 to 3:00	The public heat check on a phone, then the live URL and repo.	"Anyone can check their own spot in ten seconds." What it does not do (evidence, not proof), thanks.

- Record the product segments on Saturday night after the freeze, while everything works. Record the voice separately.
- Warm the Lambdas before recording so nothing waits on a cold start.
- Show the Replay banner on screen whenever replay data is used.

- Say the limitation yourself: public forecasts are coarse and the site factors are cited assumptions. It is more convincing than leaving a judge to spot it.

SECTION 18

Submission, risks and the pre-kickoff checklist

Risk	Mitigation
Site parameters questioned	Cited values per parameter, sensitivity analysis, cautious defaults, optional sensor check.
Judge says this already exists	Section 1 names the existing tools and states what is added. Never claim to be first; claim the gap.
Symptom buttons misused to stop work	Needs several separate phones in one break window; tighten-only; every trigger is logged for the officer.
Agent misreads a heat action plan	Exact-quote check against the page; human approval; scored against a hand-made list for two plans; approved rulebooks shipped as fallback.
Plan is scanned or not in English	Textract for scanned pages; choose two plans with usable text before Thursday.
Rest record gamed by a supervisor	Time window, one response per phone, three-response minimum; stated openly as evidence, not proof.
Workers will not scan, or fear blame	Anonymous counts only; test the idea with real workers before Thursday; shared phone works.
Re-plan looks unrealistic	Supervisor ticks which changes are allowed; the tool says when the current order is already fine.
Thresholds misread	Transcribe from NIOSH with page references; version the file; expert review; property tests.
Agent says something inconsistent	Agent never decides; numeric check; template fallback; fixed footer.
October weather shows no alert	Replay of a documented hot day, now P0.
WBGT library will not package for Lambda	Test the layer in the first hours of Thursday; fall back to a container-image Lambda if needed.
WhatsApp or Bedrock access delayed	Start both now; email, templates and the manual form are the baseline.
Over-scoping	Outline, checkpoints, Saturday freeze.
Repository history questioned	No pre-written code; commit from hour one; list AI tools; credit everything.

Submission checklist

README order, borrowed from the winners: the problem as a human moment; who it is for; what it does; try it (live link and self-playing demo); where AWS fits, one line per service; the safety model as a numbered list; run it; tests; limitations; AI tools used; credits.

Item	Requirement	Done
Public repository	History spans Oct 8 to 11; README with run steps, architecture and credits; no secrets.	[]
Demo video	Under 3 minutes, shows AWS and the working system, opens signed out.	[]
Writeup	Problem, build, where AWS fits, AI coding tools used, limitations.	[]
Credits	Open-Meteo, WBGT library, NIOSH, other libraries, any footage or assets not made by the team.	[]
Team	Every teammate registered; Builder Center profiles verified.	[]
Blog and live URL	Blog published and linked; frontend and API stay up through judging.	[]

Before Thursday (allowed: research, design and account setup, no project code)

- Finish AWS Builder Center verification for every teammate; open the AWS account; request Bedrock access for one text model and one image-capable model.
- Find and note a cited value for each site parameter (Section 5).
- Read NIOSH 2016 and mark the exact tables and pages to transcribe.

- Talk to one or two real supervisors, ideally through the HOD. Ask what they do when a heatwave SMS arrives and whether they could start a shift earlier. Ask the HOD for an expert reviewer.
- Talk to two or three workers as well: would they scan a QR to confirm a break, and what would stop them? Arrange a site, even a campus construction site, where you can run a short field test on Saturday.
- Download two heat action plans (Delhi 2026 and your own state's or city's). Read them and list every worker-facing obligation with its page number. Check the PDFs have selectable text.
- Read the official text of the Delhi Heat Wave Action Plan 2026 and your own state's plan; note the exact rest window and who it covers.
- Write down a realistic task list for one real site; it becomes the re-planner's demo input.
- Pick the pilot site and the replay hot day, and confirm the historical data exists.
- Start the Meta developer account and WhatsApp Business Account if you still want WhatsApp.
- Install Python, Node, AWS CLI and SAM CLI. Decide whether the team is two or four.

Originality rule

Do not create the repository, scaffold the project, write pipeline code, or reuse code from earlier projects before the clock starts on Oct 8. Open-source libraries, public APIs and starter templates are allowed during the event; what is judged is what you add during it.